

ISAAC ADJEI

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[Date]

[Hiring Manager Name/Recruitment Team]

[Company Name]

[Company Address/Location]

Dear [Hiring Manager Name/Recruitment Team] ,

I am writing to apply for the [Role Title] position at [Company Name] . As an Electronic Engineering and Computer Science student at Aston University (Predicted First Class) and Top 40 Finalist in the Black Heritage Undergraduate of the Year Award 2026, I bring a rigorous mathematical foundation built through four years of engineering mathematics, signal processing and control theory: and the programming discipline to apply that foundation to real quantitative problems.

My degree covers linear algebra, complex analysis, Laplace and Z-transforms, probability and stochastic processes and digital signal processing: not as isolated modules but as tools applied continuously in project work. In the PHAEMOS platform I implemented an FFT-based vibration analysis pipeline running on STM32 hardware using HAL C: sampling at 8 kHz, applying a Hanning window, computing frequency bins and flagging spectral anomalies against baseline thresholds. Translating a DSP algorithm from theory to constrained hardware, where floating-point operations are expensive and memory is tight, is an exercise in mathematical precision of the kind that transfers directly to numerical computing in financial contexts. My two-stage audio amplifier project (TL071/OPA551, custom PCB, 281mW output) required frequency response analysis, gain-bandwidth tradeoffs and stability margin calculations, all verified against Proteus simulation data. Beyond hardware I have applied unsupervised machine learning (Isolation Forest) to 25-field IoT telemetry, engineered rolling statistical features in Python and run numerical pipelines in numpy and pandas. I am drawn to [Company Name] because of [specific reason] and I am eager to contribute to [specific quantitative or analytical challenge] .

Quantitative roles reward the ability to work carefully under uncertainty: to distinguish signal from noise, to question whether a model's assumptions match the data and to communicate results precisely. My engineering training teaches all three. I am comfortable moving between mathematical abstraction and working code, and I treat correctness as a constraint rather than a goal: a programme is not done when it runs, it is done when I can prove it is right on the cases that matter.

I am actively developing my quantitative skills alongside my degree: studying competitive programming on Leetcode and Codeforces, working through DSP and numerical methods problems and applying statistical reasoning to my own data pipelines. [Company Name] 's work on [specific quantitative domain or tool] is the environment where I want to grow, and I would bring intellectual rigour, precise implementation and a genuine appetite for the mathematical depth the role requires.

Thank you for considering my application. I would welcome a conversation about how my mathematical and engineering background can contribute to [Company Name]'s quantitative work.

Yours sincerely,

Isaac Adjei